

TitanOS

The Operating System for Personal AGI.

TitanOS is an open-source AI Operating System designed to give artificial intelligence something modern LLMs fundamentally lack:

Memory. Learning. Growth.

Instead of treating AI as a stateless chatbot, TitanOS treats AI as a continuously evolving digital mind capable of:

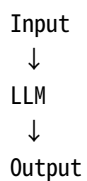
- Long-term memory
- Knowledge organization
- Goal planning
- Autonomous execution
- Experience accumulation
- Digital twin creation

Our vision is simple:

Build an AI that grows with its user for years.

Why TitanOS?

Today's AI systems suffer from a fundamental limitation:



```
graph TD; Input --> LLM; LLM --> Output;
```

Input
↓
LLM
↓
Output

After the conversation ends:

- knowledge disappears
- experiences disappear
- learning disappears

The AI starts over.

TitanOS introduces persistent cognition.



```
graph TD; User --> Box[ ];
```

User
↓

Perception
↓
Memory
↓
Reasoning
↓
Planning
↓
Execution
↓
Learning

The system continuously improves through interaction.

Vision

TitanOS aims to become:

Linux for Personal AGI.

An open infrastructure where developers can build intelligent agents that:

- remember
- reason
- learn
- evolve

over years rather than minutes.

Core Architecture

TitanOS

├── Gateway
├── Brain
├── Memory Engine
├── Knowledge Graph
├── Planner
├── Executor
├── Learning Engine
├── Skill Store
└── Digital Twin

Features

Long-Term Memory

Store and retrieve:

- conversations
- documents
- experiences
- goals
- preferences
- learned knowledge

Memory is ranked by:

importance
frequency
recency

Similar to how human memory works.

Knowledge Graph

Convert raw information into structured relationships.

Example:

User
↓ Likes
Machine Learning

Machine Learning
↓ belongs_to
Artificial Intelligence

Benefits:

- relationship reasoning
 - contextual understanding
 - personalized recommendations
-

Autonomous Planner

Transform goals into actionable plans.

Example:

Input:

I want to become a machine learning engineer.

Output:

1. Learn Python
2. Learn Data Structures
3. Learn Statistics
4. Learn Machine Learning
5. Build Projects
6. Apply for Jobs

Skill Store

Install and execute skills dynamically.

Examples:

- Web Search
- Coding
- Research
- Data Analysis
- Automation
- Document Processing

Developers can publish their own skills.

Learning Engine

TitanOS learns from outcomes.

Every completed task generates experience:

```
{  
  "task": "Build crawler",  
  "result": "Success",  
  "duration": "12min",  
  "lesson": "Check anti-bot mechanisms first"  
}
```

These experiences become reusable knowledge.

Digital Twin

Create an AI representation of a user.

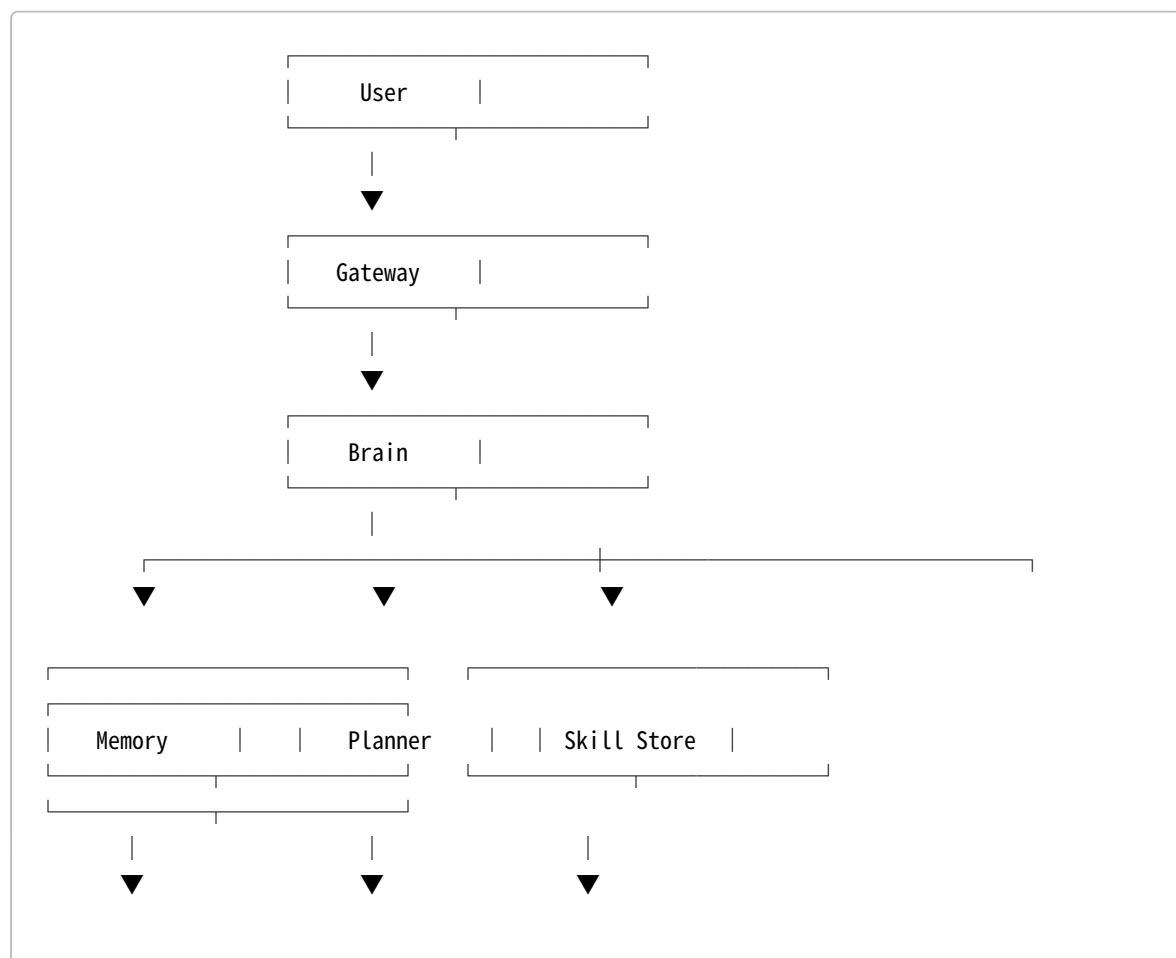
The Digital Twin gradually learns:

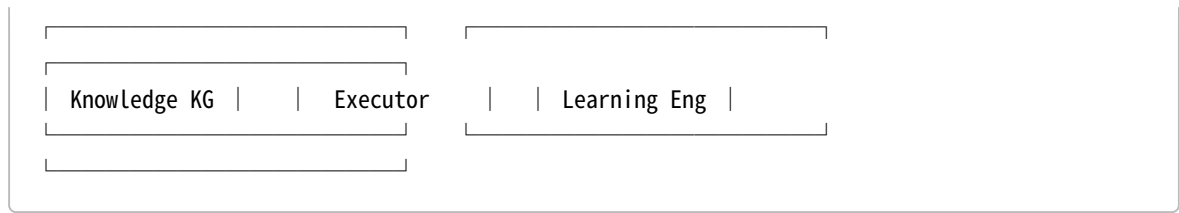
- writing style
- coding style
- decision patterns
- interests
- habits

Goal:

You
↓
AI Version of You

System Design





Technology Stack

Backend

- Python
- FastAPI
- PostgreSQL
- Redis

AI Layer

- Qwen
- DeepSeek
- Llama
- OpenAI Compatible APIs

Memory Layer

- Milvus
- Weaviate
- Chroma

Knowledge Graph

- Neo4j

Frontend

- Next.js
- TypeScript
- TailwindCSS

Deployment

- Docker
 - Kubernetes
-

Roadmap

v0.1

- Chat Interface
- Memory Storage
- Vector Search
- Basic RAG

v0.5

- Planner
- Agent Execution
- Tool Calling
- Skill Store

v1.0

- Long-Term Learning
- Experience Memory
- Personalized Memory Ranking

v2.0

- Digital Twin
- Personal Knowledge Graph
- Behavior Modeling

v3.0

- Self-Evolving Personal AGI

Repository Structure

TitanOS/

backend/

|

|— app/

|

|— api/

|

|— brain/

|

|— memory/

|

|— planner/

|

|— executor/

|

|— learning/

|

|— skills/

```
|   └── digital_twin/
|
|── database/
|
|── tests/
|
|── configs/

frontend/
|
|── pages/
|── components/
|── hooks/
|── services/

docs/
|
|── architecture/
|── api/
|── roadmap/
```

Quick Start

```
git clone https://github.com/yourname/TitanOS.git

cd TitanOS

docker compose up -d
```

Backend:

```
uvicorn app.main:app --reload
```

Frontend:

```
npm install

npm run dev
```


Research Directions

TitanOS explores:

- Long-Term Memory Systems
 - Agent Architectures
 - Knowledge Graph Reasoning
 - Experience-Based Learning
 - Autonomous Planning
 - Digital Human Modeling
 - Personal AGI
-

Contributing

We welcome:

- AI researchers
- Backend engineers
- Frontend engineers
- System architects
- Knowledge graph experts
- Agent developers

Please read:

CONTRIBUTING.md

before submitting pull requests.

License

Apache License 2.0

Philosophy

Most AI systems answer questions.

TitanOS aims to build something different.

An intelligence that remembers.

An intelligence that learns.

An intelligence that grows.

The future of AI is not bigger models.

The future of AI is persistent intelligence.